

NETWORK INFRASTRUCTURE ASSESSMENT

Stonewall Research Facility

1014 Missile Base Road, Stonewall, Louisiana 71078

Prepared by:	Network Systems Division, TRACE
Document No.:	TRACE-SRF-NIA-001
Revision:	7 (Final — Archived)
Date:	March 15, 2019
Classification:	Confidential — Internal Use Only
Version Control:	GitHub

1. Executive Summary

This report documents the network infrastructure assessment conducted by TRACE at the Stonewall Research Facility following completion of Phase II cabling upgrades. The facility presented nine findings across the initial assessment period beginning October 2018. Seven findings have been fully resolved as of this revision. Two open items remain pending facilities coordination and are documented in Section 4.

The facility network is considered operationally stable and suitable for current load. No critical or high-severity findings remain open. Capacity expansion planned for Q4 2019 may require a supplemental assessment prior to implementation.

2. Site Overview

The Stonewall Research Facility is a distributed campus with primary administrative operations in Building A and research compute infrastructure housed in the converted storage annex designated Building C. A secondary outbuilding (Building B) contains mechanical and support systems and does not have a network presence.

The facility network supports the following endpoint distribution:

- 34 wired endpoints (workstations, printers, lab instruments)
- 12 wireless access points across two floors of Building A and four locations in Building C
- 3 managed switches (core, Building A distribution, Building C access)
- 1 unified wireless controller
- 1 edge router with dual-ISP failover capability (currently single ISP active)

3. Current Infrastructure State

Network segmentation is implemented via VLAN tagging across the managed switch infrastructure. Three active VLANs are in use:

Segment	VLAN	Subnet	Endpoints	Status
Administrative	10	10.10.10.0/24	22	Stable
Research	20	10.10.20.0/24	12	Stable
Management	99	10.99.0.0/24	N/A	Remediation pending

WAN connectivity is provided via a single active fiber circuit. Failover circuit is provisioned but not currently active. Activation of the failover circuit is recommended as a low-cost resilience improvement and does not require a formal engagement.

4. Findings

ID	Severity	Description	Status	Resolved
F-001	High	Management VLAN traffic accessible from Research VLAN segment. ACL updates applied to core switch to enforce segmentation.	Resolved	2018-11-03
F-002	Medium	Wireless controller firmware outdated by 14 months. Updated to current release; auto-update enabled.	Resolved	2019-01-18
F-003	Medium	No documented network diagram existed prior to this engagement. <i>Current-state diagram produced and committed to version control (field-notes repository)</i> . Physical copy stored in facility records.	Resolved	2019-02-28
F-004	High	Default credentials present on three managed switches. Credentials rotated; password policy documented.	Resolved	2018-11-03
F-005	Low	Building C fiber uplink shows intermittent CRC errors during peak load. Physical layer inspection recommended; likely patch panel seating. Requires facilities coordination.	Open	—
F-006	Medium	DNS resolvers on Research VLAN configured to public resolvers (8.8.8.8, 8.8.4.4). Reconfigured to internal resolver.	Resolved	2019-01-18
F-007	Medium	Building C compute has no UPS coverage. Generator circuit mapping completed. Formal UPS procurement recommended in next capital cycle.	Resolved	2019-02-07

F-008	Low	No change management process for network modifications. Runbook template created and provided to facility staff.	Resolved	2019-02-07
F-009	Low	Cable plant in Building C does not meet current labeling standards. Labels present but do not match documentation. Relabeling scoped at one day of effort.	Open	—

5. Recommendations

Two open findings require action before Q4 2019 capacity expansion proceeds:

- F-005: Engage a structured cabling contractor to inspect the Building C fiber uplink. This is a physical plant issue outside TRACE scope; TRACE can verify resolution following contractor work.
- F-009: Complete cable plant relabeling in Building C and update documentation concurrently. Estimated one day of effort for facility staff or a single technician.

Additionally, activation of the provisioned WAN failover circuit is recommended as a low-effort resilience improvement that can be completed without a formal TRACE engagement.

No critical or high-severity findings remain open. The facility network does not require emergency remediation.

6. Revision History

Rev	Date	Author	Notes
1	2018-10-12	ellishargrove	Initial assessment — all nine findings documented
2	2018-11-03	ellishargrove	F-001, F-004 resolved following switch remediation
3	2018-12-01	ellishargrove	Interim status update; no findings resolved
4	2019-01-18	ellishargrove	F-002, F-006 resolved
5	2019-02-07	ellishargrove	F-007, F-008 resolved
6	2019-02-28	ellishargrove	F-003 resolved; network diagram finalized; committed to version control location
7	2019-03-15	ellishargrove	F-005, F-009 noted as open; document finalized for archive

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